Acitivity:

1. Medical Image Enhancement

1) Gamma (power-Law) Transformation with $\gamma < 1$

2) • It enhances the dark regions of image
• It improves the visibility of Sof tissues without significantly affecting the already bright bone regions

3) • Dark tissues become brighter and clearer.
• More hidden details in low-intensity regions become visible
• Overall image quality and diagnostic usefulness improve

2) Night-Time Surveillance

1) Log Transformation.

2) * Log Transformation expands low-intensity pixel values, making dark areas brighter.
* Negative transformation only inverts image

3) * The parking lot appears brighter.
* Vehicles, people and other objects become easier to identify
* Details hidden in shadows become more visible while bright areas remain unchange

3) Fingerprint Recognition

1) Contrast Streching (Linear Contrast Enhancement)

2) * It increases the difference between the ridge and valley intensities.

* Fine fingerprint patterns become Sharper and Easier to detect.

* It improves recognition accuracy.

3) * Log transformation mainly brightens dark regions

* It doesnot effecitively increase the Contrast between Ridges & Valleys